Ripudaman Singh

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Summary

Software Developer focused on building high-velocity backend systems, deploying cloud-native infrastructure, and optimizing pipeline metrics. Experienced in navigating fast-growing startup ambiguity to optimize product reliability and streamline developer workflows.

Education

Memorial University of Newfoundland | *St. John's, NL* Jan. 2021 – Feb. 2025
Bachelor of Science in Computer Science (Honours) – GPA: 4.0

Technical Skills

Languages: TypeScript, Python, C/C++, SQL, Go
Frameworks: React, Vue, Flask, ExpressJS, FastAPI, Hono, SQLAlchemy, Drizzle
Infrastructure: SST, AWS, Docker, Kubernetes, GitHub Actions
Observability: Grafana LGTM Stack (Alloy, Mimir, Loki, Tempo), Langfuse, Posthog, Sentry, Mixpanel

Experience

trophi.ai | *St. John's, NL* Jan. 2026 – Jun. 2026
Software Developer (Full Time)
- Deployed scalable, cloud-native infrastructure using SST and AWS, driving initiatives to optimize platform reliability and minimize downtime.
- Designed and delivered an end-to-end, real-time telemetry coaching pipeline, integrating RAGFlow and Langfuse for deep context retrieval, LLM observability, and automated evaluation.
- Instrumented and deployed the Grafana LGTM stack to centralize infrastructure metrics, streamlining alerting and accelerating developer debugging velocity.

Visual and Analytic Computing Lab (Memorial University) | *St. John's, NL* May 2024 – Dec. 2024
Research Assistant (Full Time)
- Designed and conducted a user study for 30 participants.
- Developed data clean-up, aggregation and analysis solution for the user study using Pandas.

Visual and Analytic Computing Lab (Memorial University) | *St. John's, NL* May 2023 – Dec. 2023
Software Developer (CO-OP, Full Time)
- Developed a flight path visualization XR application in Unreal Engine 5 for Microsoft Hololens 2.
- Implemented interfacing using UMG and the ability to manipulate flight paths through hand tracking.

Data and Image Analysis Group (Memorial University) | *St. John's, NL* Jan. 2023 – Apr. 2023
Data Quality Assurance Analyst
- Developed an SSH manager utility using Python to help setup systems seamlessly.
- Contributed to a scheduler for a HPC payload processing system.

Department of Science (Memorial University) | *St. John's, NL* Sep. 2022 – Dec. 2022
Teaching Assistant (Computer Science)
- Assisted first-year students with core computer science concepts such as DFAs, Turing Machines and Regular Expressions.
- Graded course work, conducted lab sessions and provided feedback for 40+ students.

Projects

MediSync AI | *ExpressJS, Flask, MongoDB, OpenAI API, React, RAG* May 2024
- Won Best Overall at the DO.IT Hackathon as an aide to healthcare professionals, facilitating them to perform CRUD operations using natural language and without the need for technical experience.

Honours Thesis Project | *Python, matplotlib, opencv, 3D Slicer, tobii* Dec. 2024
- Developed a framework for collecting and visualizing eye-tracking data for medical image analysis.
- Automated loading of DICOM images and ROIs in 3D Slicer and visualizing the spatial data using heatmaps.

Bubblegum Bloodbath (C++ Game) | *C++, SFML, ImGUI* Dec. 2024
- Developed a C++ game from scratch using SFML as part of a group project.
- Won Best Overall in the batch.

Car Rental Full Stack | *TypeScript, React, Chakra UI, Node.js, Express, MySQL* Dec. 2024
- A full-stack car rental application with user authentication, vehicle management, and booking system.